

Computer Networks



Module 4: SNMP, ASN.1, Physical Layer

- SNMP
- ASN.1
- Physical Layer:
 - Data and signals
 - Digital transmission
 - Analog transmission
 - Bandwidth utilization
 - Transmission media

7.2 Digital Transmission

Basic Idea

Information must be converted into a signal before transmission.

Digital Data $\rightarrow$ Digital Signal

Analog Data $\rightarrow$ Digital Data

Topics

- 1 Digital-to-Digital Conversion
- 2 Analog-to-Digital Conversion

Digital-to-Digital Conversion

Digital data can be represented using digital signals.

Three Techniques

- ① **Line Coding** – always needed
- ② **Block Coding** – may be needed
- ③ **Scrambling** – may be needed

Main Idea

Digital data → digital signal.

Line Coding

Definition

Line coding is the process of converting digital data into digital signals.

Digital Data $\xrightarrow{\text{Encoding}}$ Digital Signal

Digital Signal $\xrightarrow{\text{Decoding}}$ Digital Data

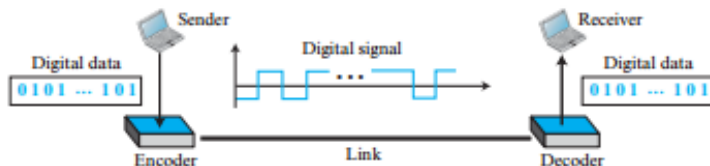
$$S_{avg} = c \frac{N}{r}$$

Parameters

- N = bit rate
- r = bits carried by each signal change
- S = baud rate
- S_{avg} = average signal changes

Figure 7.19 – Line Coding

Figure 7.19 *Line coding and decoding*



Polar Schemes

Polar schemes use positive and negative voltage levels.

Main Polar Schemes

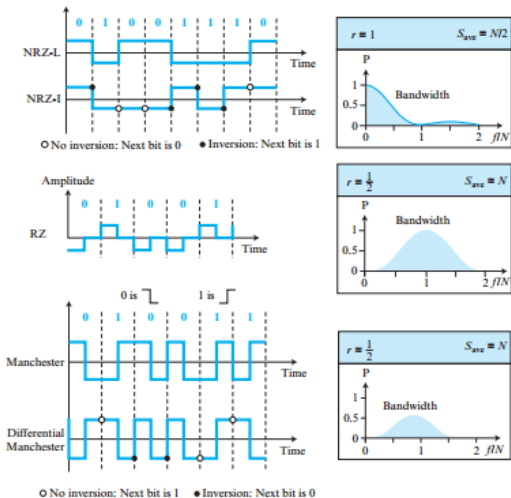
- NRZ-L
- NRZ-I
- RZ
- Manchester
- Differential Manchester

Problem

Long sequences of identical bits can create synchronization problems.

Figure 7.20 – Polar Schemes

Figure 7.20 Polar schemes



NRZ-L and NRZ-I

NRZ-L

The voltage level determines the bit.

- One voltage level represents 0.
- The other represents 1.

NRZ-I

The change or absence of change determines the bit.

- No change $\rightarrow$ 0
- Change $\rightarrow$ 1

Problem

Long sequences can skew the average power and create synchronization problems.

RZ – Return-to-Zero

RZ uses three levels:

$+$, 0 , $-$

- Signal returns to zero during each bit.
- Signal changes during the bit.
- Provides better synchronization.
- Requires greater bandwidth.

Trade-off

Better synchronization $\leftrightarrow$ greater bandwidth.

Manchester and Differential Manchester

Manchester

Combines RZ and NRZ-L.

A transition always occurs at the middle of the bit, providing synchronization.

Differential Manchester

Combines RZ and NRZ-I.

- Always has a middle transition.
- Middle transition provides synchronization.
- Beginning transition determines the bit.

Bipolar Schemes

Bipolar schemes use:

+, 0, -

AMI – Alternate Mark Inversion

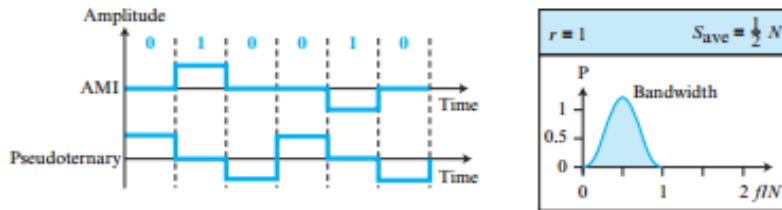
- 0 → zero voltage
- 1 → alternating positive and negative

Pseudoternary

- 1 → zero voltage
- 0 → alternating positive and negative

Figure 7.21 – Bipolar Schemes

Figure 7.21 Bipolar schemes: AMI and pseudoternary



Multilevel Schemes

Purpose

Multilevel schemes are designed to increase data speed or decrease bandwidth.

m data elements $\rightarrow n$ signal elements

Notation:



where:

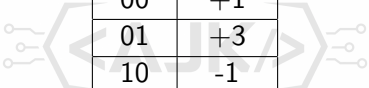
- m = binary pattern length
- n = signal pattern length
- L = number of signal levels

Examples:

$2B1Q$

$8B6T$

Two bits are encoded into one four-level signal element.



Bits	Level
00	+1
01	+3
10	-1
11	-3

If the previous signal level is negative, the sign can be inverted according to the coding rule.

8 bits $\rightarrow$ 6 ternary elements

Number of data patterns:

$$2^8 = 256$$

Number of signal patterns:

$$3^6 = 729$$

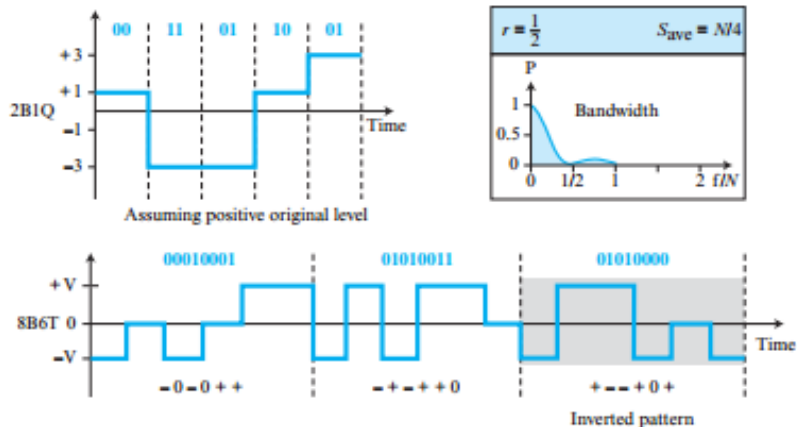
Redundant patterns:

$$729 - 256 = 473$$

Redundancy provides synchronization and error-detection possibilities.

Figure 7.22 – Multilevel Schemes

Figure 7.22 Multilevel: 2B1Q and 8B6T



Block Coding

Block coding adds redundancy for:

- Synchronization
- Inherent error detection

$$mB/nB, \quad n > m$$

Three Steps

Division $\rightarrow$ Substitution $\rightarrow$ Combination

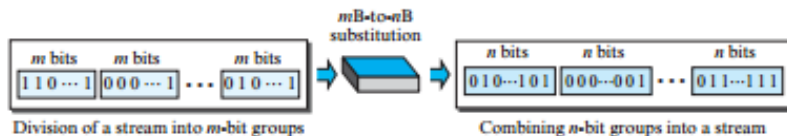
Examples:

$$4B/5B$$

$$8B/10B$$

Figure 7.23 – Block Coding

Figure 7.23 Block coding concept



4 bits $\rightarrow$ 5 bits

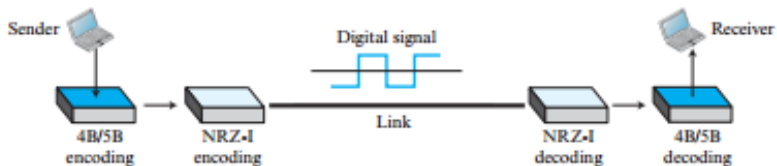
- Designed to work with NRZ-I.
- Prevents long sequences of zeros.
- Encoded stream contains no more than three consecutive zeros.
- Receiver removes the redundant bits.

Purpose

Maintain synchronization.

Figure 7.24 – 4B/5B

Figure 7.24 *Using block coding 4B/5B with NRZ-I line coding scheme*



$8 \text{ bits} \rightarrow 10 \text{ bits}$

It combines:

$$5B/6B + 3B/4B$$

Redundant groups:

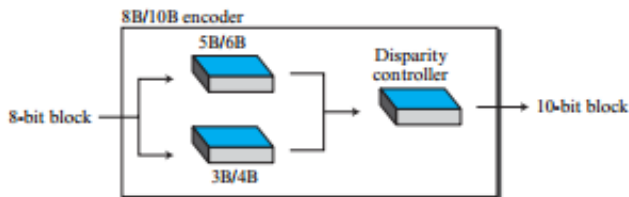
$$2^{10} - 2^8 = 1024 - 256 = 768$$

Purpose

Provides synchronization and error detection.

Figure 7.25 – 8B/10B

Figure 7.25 8B/10B block encoding



Scrambling

Problem with AMI

AMI has narrow bandwidth and no DC component, but long sequences of zeros cause synchronization problems.

Solution

Scrambling replaces long zero sequences with special patterns without increasing the number of bits.

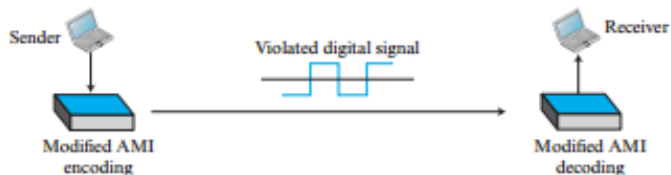
Common techniques:

B8ZS

HDB3

Figure 7.26 – Scrambling

Figure 7.26 *AMI used with scrambling*



B8ZS is commonly used in North America.

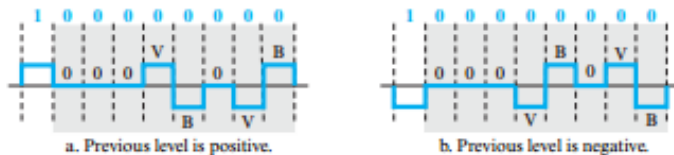
Eight consecutive zeros are replaced by:

000*V*B0*V*B

- V = violation
- B = bipolar pulse
- Bit rate remains unchanged.
- Positive and negative pulses remain balanced.
- DC balance is maintained.

Figure 7.27 – B8ZS

Figure 7.27 Two cases of B8ZS scrambling technique



a. Previous level is positive.

b. Previous level is negative.

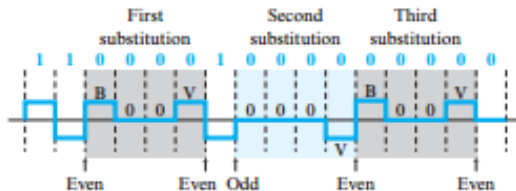
HDB3 is commonly used outside North America.
Four consecutive zeros are replaced by:

$000V$ or $B00V$

- Odd number of nonzero pulses $\rightarrow 000V$
- Even number of nonzero pulses $\rightarrow B00V$
- Maintains an even number of nonzero pulses.

Figure 7.28 – HDB3

Figure 7.28 *Different situations in HDB3 scrambling technique*



7.2.2 Analog-to-Digital Conversion

Analog sources include:

- Microphone
- Camera

Digital representation is less susceptible to noise.

Two techniques:

PCM

Delta Modulation

After digitization, digital-to-digital conversion can be used.

Pulse Code Modulation – PCM

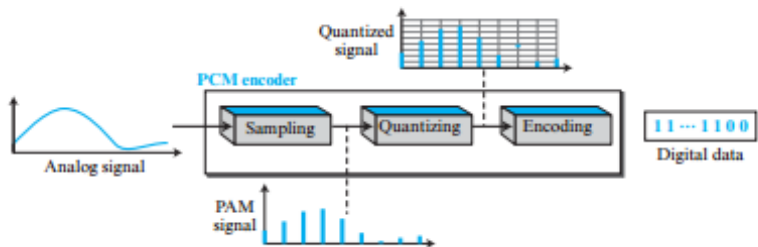
PCM is the most common analog-to-digital conversion technique.

Sampling → Quantization → Encoding

- 1 Sampling
- 2 Quantization
- 3 Encoding

Figure 7.29 – PCM

Figure 7.29 Components of PCM encoder

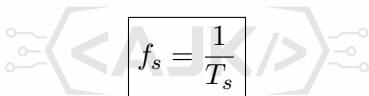


Sampling

Sampling interval:

$$T_s$$

Sampling frequency:

A diagram showing a central rectangular box containing the equation $f_s = \frac{1}{T_s}$. This box is flanked by two large, light-gray chevron symbols pointing outwards. Each chevron has three small circles at its ends, resembling a circuit board trace or a signal path.
$$f_s = \frac{1}{T_s}$$

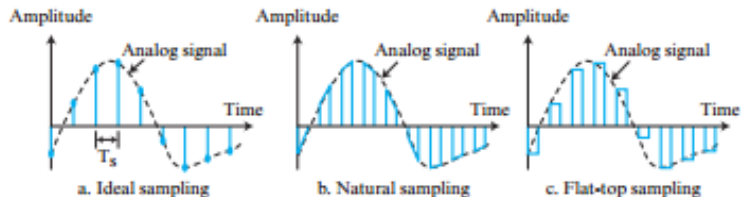
Three methods:

- 1 Ideal sampling
- 2 Natural sampling
- 3 Flat-top sampling

Sample-and-hold is the most common method.

Figure 7.30 – Sampling Methods

Figure 7.30 Three different sampling methods for PCM



Nyquist Sampling Theorem

$$f_s \geq 2f_{\max}$$

The signal must be band-limited.

Important

The sampling rate must be at least twice the highest frequency, not twice the bandwidth.

For a low-pass signal:

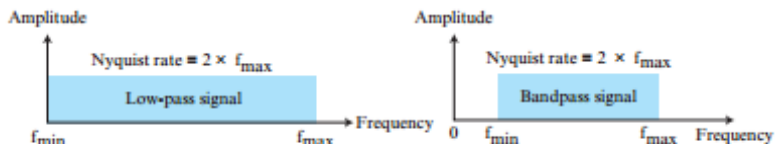
$$B = f_{\max}$$

For a band-pass signal:

$$B < f_{\max}$$

Figure 7.31 – Nyquist Sampling

Figure 7.31 *Nyquist sampling rate for low-pass and bandpass signals*



Quantization

If the signal ranges from $V_{\min}$ to $V_{\max}$ and has L levels:

$$\Delta = \frac{V_{\max} - V_{\min}}{L}$$

- 1 Divide the amplitude range into L zones.
- 2 Assign values to the midpoints.
- 3 Approximate each sample to the quantized value.

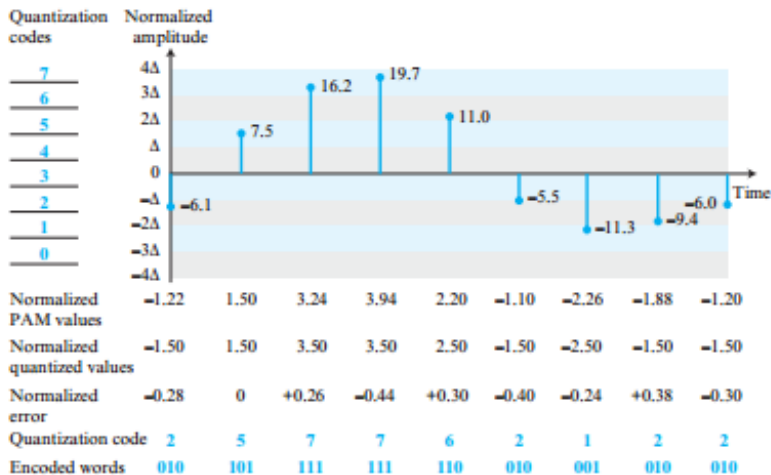
Example

For $-20V$ to $+20V$ and $L = 8$:

$$\Delta = \frac{40}{8} = \boxed{5V}$$

Figure 7.32 – Quantization and Encoding

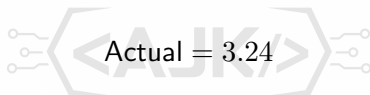
Figure 7.32 Quantization and encoding of a sampled signal



Quantization Error

$$-\frac{\Delta}{2} \leq \text{error} \leq \frac{\Delta}{2}$$

Example:



Actual = 3.24

Quantized = 3.50

$$\text{Error} = 3.50 - 3.24 = +0.26$$

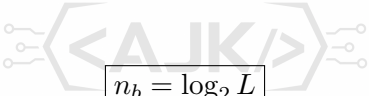
Quantization error affects SNR and therefore Shannon capacity.

Quantization Levels and SNR

More quantization levels provide greater accuracy.

- Audio normally uses $L = 256$.
- Video may use thousands of levels.

Bits per sample:


$$n_b = \log_2 L$$

Quantization SNR:

$$SNR_{dB} = 6.02n_b + 1.76$$

PCM Encoding

Each quantized sample becomes an n_b -bit code word.

$$n_b = \log_2 L$$

For $L = 8$:

Examples:

$$n_b = 3$$

$$2 \rightarrow 010$$

$$5 \rightarrow 101$$

Bit rate:

$$BitRate = f_s \times n_b$$

Example 7.13 – Human Voice

Human voice:

$$0 - 4000 \text{ Hz}$$

Therefore:

Nyquist sampling rate:


$$f_{\max} = 4000 \text{ Hz}$$
$$f_s = 2(4000) = 8000$$

$$f_s = 8000 \text{ samples/s}$$

With 8 bits/sample:

$$\text{BitRate} = 8000 \times 8$$

$$\text{BitRate} = 64 \text{ kbps}$$

Original Signal Recovery

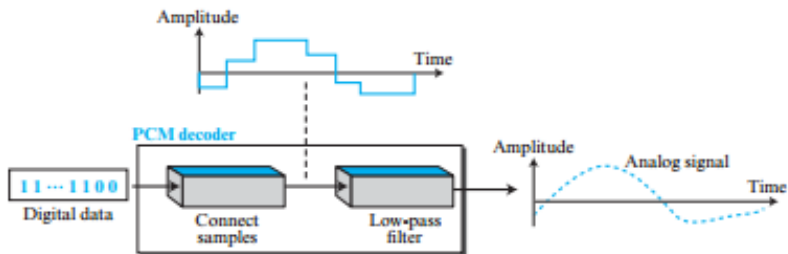
PCM decoder converts code words into pulses.

Code Words $\rightarrow$ Pulses $\rightarrow$ Staircase Signal $\rightarrow$ Low-Pass Filter $\rightarrow$ Analog Signal

If the signal is sampled at or above the Nyquist rate and sufficient quantization levels are used, the original signal can be recreated.

Figure 7.33 – PCM Decoder

Figure 7.33 Components of a PCM decoder



PCM Bandwidth

Minimum digital bandwidth:

$$B_{\min} = n_b B_{\text{analog}}$$

Thus, digital bandwidth is n_b times the analog bandwidth.

Price of Digitization

Digitization requires additional digital bandwidth.

$$\text{BitRate} = f_s \times n_b$$

Delta Modulation

Delta Modulation is simpler than PCM.

PCM

Finds the amplitude of each sample and sends a code word.

DM

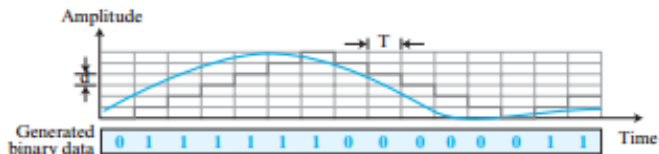
Finds the change from the previous sample and sends bits one after another.

$$PCM = \text{sample value}$$

$$DM = \text{change between samples}$$

Figure 7.34 – Delta Modulation

Figure 7.34 The process of delta modulation



Section 7.2 – Quick Revision

Digital-to-Digital

- Line Coding
- Block Coding
- Scrambling

Analog-to-Digital

- Sampling
- Quantization
- Encoding
- PCM
- Delta Modulation

Important Formulas

$$S_{avg} = c \frac{N}{r}$$

$$f_s = \frac{1}{T_s}$$

$$f_s \geq 2f_{\max}$$

$$\Delta = \frac{V_{\max} - V_{\min}}{L}$$

$$n_b = \log_2 L$$

$$BitRate = f_s n_b$$

$$SNR_{dB} = 6.02n_b + 1.76$$

Important Exam Points

- 1 NRZ-L versus NRZ-I
- 2 RZ and Manchester synchronization
- 3 Manchester versus Differential Manchester
- 4 AMI versus Pseudoternary
- 5 2B1Q and 8B6T
- 6 4B/5B and 8B/10B
- 7 Need for scrambling
- 8 B8ZS and HDB3
- 9 Nyquist sampling theorem
- 10 Quantization error
- 11 PCM encoding
- 12 PCM versus Delta Modulation

